

Once a Glacier

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Abstract

Once a Glacier tells the story of a relationship between a girl and a glacier. As the girl grows older, the existence of the ice is threatened, and the viewer is taken on a journey through her seemingly futile efforts to protect what was once an entire glacier. The story is inspired by Jiabao Li's own experience from two years that she lived in Alaska, where she was the young girl protecting a piece of ice in the freezer. In Inupiaq tradition, a tribe of indigenous people from northwestern Alaska, glaciers carry memories from the past and communicate them in song. The climate crisis has become a terrifying reality that includes seeing the end of glaciers—the end of these sung histories—happen before our eyes. *Once a Glacier* shows a gesture of nurturing, and whether the girl's efforts are successful or not, the work suggests a poignant level of grace and humility for moving forward into the future.

CCS Concepts

• Arts and Humanities; • Multimedia Information Systems and Virtual Reality; • Three-Dimensional Graphics and Realism;

Keywords

embodied experience, climate change, glacier, virtual reality

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1 Introduction

Once a Glacier tells the story of a relationship between a girl and a glacier. As the girl grows older, the existence of the ice is threatened, and the viewer is taken on a journey through her seemingly futile efforts to protect what was once an entire glacier. The story is inspired by Jiabao Li's own experience from two years she lived in Alaska, where she was the young girl protecting a piece of ice in the freezer. In Inupiaq tradition, a tribe of indigenous people from northwestern Alaska, glaciers carry memories from the past and communicate them in song. The climate crisis has become a terrifying reality that includes seeing the end of glaciers—the end of

these sung histories—happen before our eyes. *Once a Glacier* shows a gesture of nurturing, and whether the girl's efforts are successful, the work suggests a poignant level of grace and humility for moving forward into the future.

The team recorded the sound on the glaciers in Alaska and modeled the glaciers from the satellite data of Matanuska glaciers, to document the disappearing nature. The VR film relates the life of a human to natural phenomena. We think of nature's time scale as millions of years and humans' are under a hundred, yet with the current speed of climate change, many glaciers are vanishing within our lifetime. Glacial time, once was slow, is now fast. By telling the story of the life journey of a piece of glacial ice and its glacier of origin along with the life of a girl, the film makes this human-nature timescale tangible.

2 Exhibition setup

This VR theater entry can be configured in various formats, depending on the site requirements. We have two versions: a 360 film and an interactive PCVR film. In the interactive version, the girl enlists the viewer's help to collect the ingredients and water them to grow the glacier. *Once a Glacier* is a multisensory experience. Besides the soundscape from real glaciers in Alaska, we will have a scent depositor to shoot the scent of glacier, swamp, freezer, and soggy museum at certain time as the story evolves. The film has received critical acclaim at SXSW, IDFA, and Raindance, was nominated for Best Immersive Experience, Best Immersive Experience for Social Impact, and won the Best VR Short award at Cannes World Film Festival.

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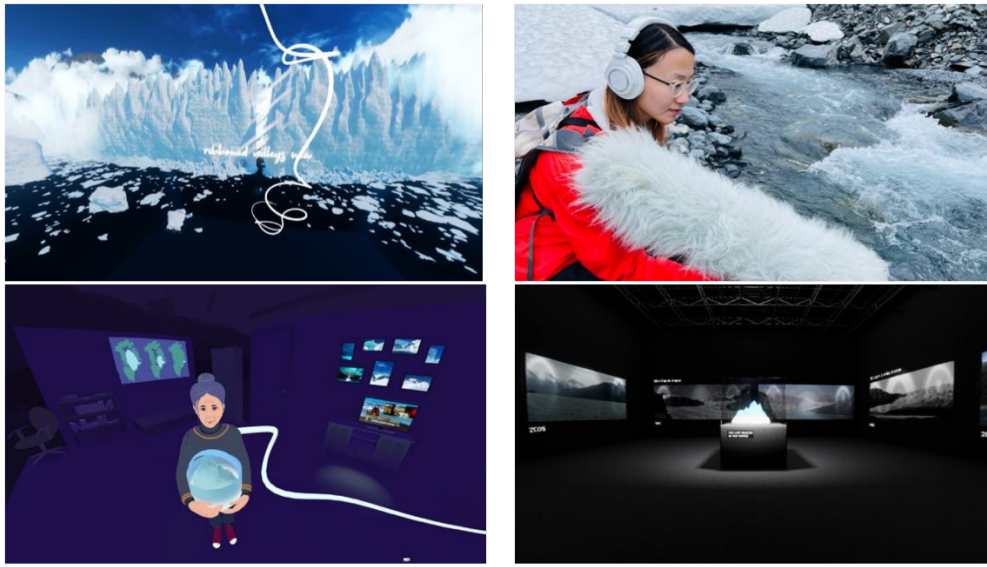


Figure 1: The glaciers are modeled from the satellite data of Matanuska glaciers, to document the disappearing nature (top left). While we were recording the sound in Alaska, I wasn't able to find the word in English to describe this chirping sound you just heard. But I found it in the indigenous record - Tagish means "the sound of the break-up of ice" (top right). We worked closely with the indigenous people of Alaska. Inupiaq poet Joan Kane wrote the poem that became the grandma Carolyn Nahyoumaurak's voice over. (bottom left). The last scene, the glacier ice has been auctioned and placed in a museum glass box. We referenced the before and after images from the NASA data base (bottom right).